How do we evaluate Uncertainty Estimation Methods?



Evaluating Uncertainty Estimation

How to evaluate the quality of uncertainty estimates?

- Proper Scoring Rule (LL, Brier Score)
  - [Gneiting and Raftery, "Strictly Proper Scoring Rules, Prediction, and Estimation", JASA 2007]
- Calibration (NLL, ECE)
  - ["Obtaining Well Calibrated Probabilities Using Bayesian Binning", AAAI 2015]
- Out-of-distribution detection

